

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EC-7004(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Power Electronics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions
ii) All questions carry equal marks.

1. a) Explain the working principle of SCR with two transistor analogy, also draw its circuit diagram. 7
b) Explain various turn-on methods of SCR. 7
2. a) i) What do you understand by fast recovery diodes? 3
ii) Differentiate between power MOSFETs and power transistors. 4
b) An SCR is used as a static switch to turn on current in the highly inductive circuit as given in fig.1, the latching current of the SCR is 250mA. What is the minimum duration of the gate drive pulse required to turn on the SCR switch? 7

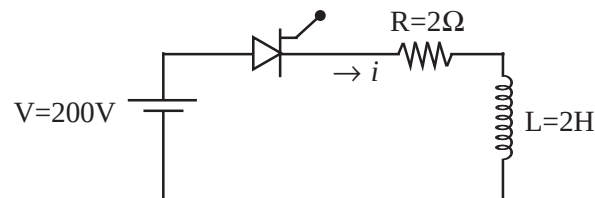


Figure 1

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3. a) Compare single phase half and full wave rectifiers. 7
b) Compare controlled and uncontrolled single phase rectifier. Draw the circuit for half wave rectifier using SCR. 7
4. a) Discuss about the effect of freewheeling diodes and source inductance on rectifier performance. 7
b) Compare McMurray and McMurray bedford inverters. 7
5. a) What do you understand by Harmonics? Explain about how harmonics can be minimized. 7
b) Explain the working principle of self cumulated inverters. 7
6. a) Compare midpoint and bridge rectifier circuits. 7
b) Explain about various control strategies in chopper. 7
7. Discuss the working principle of switching regulators 14
i) Boost
ii) Circuit regulators, along with their circuit diagrams.
8. Write short notes on any two of the following: 14
a) Current and voltage commutation of chopper
b) Protection of SCR over voltage
c) Snubber circuit
d) Three phase cyclic convertors

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